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BCT 5 - Own notes

block chain technology (Jawaharlal Nehru Technological University, Kakinada)

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UNIT V

Syllabus: Technical challenges, Business model challenges, Scandals and Public perception, Government Regulations, Uses of Block chain in E-Governance, Land Registration, Medical Information Systems.

Technical challenges:

A number of technical challenges related to the blockchain, whether a specific one or the model in general, have been identified. The issues are in clear sight of developers, with different answers to the challenges posited, and avid discussion and coding of potential solutions. Some think that the de facto standard will be the Bitcoin blockchain, as it is the incumbent, with the most widely deployed infrastructure and such network effects that it cannot help but be the standardized base. Others are building different new and separate blockchains (like Ethereum) or technology that does not use a blockchain (like Ripple). One central challenge with the underlying Bitcoin technology is scaling up from the current maximum limit of 7 transactions per second (the VISA credit card processing network routinely handles 2,000 transactions per second and can accommodate peak volumes of 10,000 transactions per second), especially if there were to be mainstream adoption of Bitcoin.

Throughput: The Bitcoin network has a potential issue with throughput in that it is processing only one transaction per second (tps), with a theoretical current maximum of 7 tps. Core developers maintain that this limit can be raised when it becomes necessary. One way that Bitcoin could handle higher throughput is if each block were bigger, though right now that leads to other issues with regard to size and blockchain bloat. Comparison metrics in other transaction processing networks are VISA (2,000 tps typical; 10,000 tps peak), Twitter (5,000 tps typical; 15,000 tps peak), and advertising networks (>100,000 tps typical).

Latency: Each Bitcoin transaction block takes 10 minutes to process, meaning that it can take at least 10 minutes for your transaction to be confirmed. For sufficient security, you should wait more time—about an hour—and for larger transfer amounts it needs to be even longer, because it must outweigh the cost of a double-spend attack (in which Bitcoins are double-spent in a separate transaction before the merchant can confirm their reception in what appears to be the intended transaction).

Size and bandwidth: The blockchain is 25 GB, and grew by 14 GB in the last year. So it already takes a long time to download (e.g., 1 day). If throughput were to increase by a factor of 2,000 to VISA standards, for example, that would be 1.42 PB/year or 3.9 GB/day. The Bitcoin community calls the size problem "bloat," but that assumes that we want a small blockchain; however, to really scale to mainstream use, the blockchain would need to be big, just more efficiently accessed. This motivates centralization, because it takes resources to run the full node, and only about 7,000 servers worldwide do in fact run full Bitcoind nodes,

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meaning the Bitcoin daemon (the full Bitcoin node running in the background). Although 25 GB of data is trivial in many areas of the modern “big data” era and data-intensive science with terabytes of data being the standard, this data can be compressed, whereas the blockchain cannot for security and accessibility reasons.

Security: There are some potential security issues with the Bitcoin blockchain. The most worrisome is the possibility of a 51-percent attack, in which one mining entity could grab control of the blockchain and double-spend previously transacted coins into his own account. The issue is the centralization tendency in mining where the competition to record new transaction blocks in the blockchain has meant that only a few large mining pools control the majority of the transaction recording. At present, the incentive is for them to be good players, and some (like Ghash.io) have stated that they would not take over the network in a 51-percent attack, but the network is insecure.

Wasted resources: Mining draws an enormous amount of energy, all of it wasted. The earlier estimate cited was $15 million per day, and other estimates are higher. On one hand, it is the very wastefulness of mining that makes it trustable—that rational agents compete in an otherwise useless proof-of-work effort in hopes of the possibility of reward—but on the other hand, these spent resources have no benefit other than mining.

Usability: The API for working with Bitcoind (the full node of all code) is far less userfriendly than the current standards of other easy-to-use modern APIs, such as widely used REST APIs.

Versioning, hard forks, multiple chains: Some other technical issues have to do with the infrastructure. One issue is the proliferation of blockchains, and that with so many different blockchains in existence, it could be easy to deploy the resources to launch a 51-percent attack on smaller chains.

Some of the partial proposed solutions to the technical issues discussed here are as follows:
Offline wallets to store the majority of coins: Different manner of offline wallets could be used to store the bulk of consumer cryptocoins—for example, paper wallets, cold storage, and bit cards.

Dark pools: There could be a more granular value chain such that big crypto-exchanges operate their own internal databases of transactions, and then periodically synchronize a summary of the transactions with the blockchain—an idea borrowed from the banking industry.

Alternative hashing algorithms: Litecoin and other cryptocurrencies use scrypt, which is at least slightly faster than Bitcoin, and other hashing algorithms could be innovated.

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REST APIs: Essentially secure calls in real time, these could be used in specific cases to help usability. Many blockchain companies provide alternative wallet interfaces that have this kind of functionality, such as Blockchain.info's numerous wallet APIs.

Figure.15.1 Blockchain Technology Challenges

Business Model Challenges:

Another noted challenge, both functional and technical, is related to business models. At first traditional business models might not seem applicable to Bitcoin since the whole point of decentralized peer-to-peer models is that there are no facilitating intermediaries to take a cut/transaction fee (as in one classical business model). However, there are still many worthwhile revenue-generating products and services to provide in the new blockchain economy. Education and mainstream user-friendly tools are obvious low-hanging fruit (for example, being targeted by Coinbase, Circle Internet Financial, and Xapo), as is improving the efficiency of the entire worldwide existing banking and finance infrastructure like Ripple—another almost "no brainer" project, when blockchain principles are understood. Looking ahead, reconfiguring all of business and commerce with smart contracts in the Bitcoin 2.0 era could likely be complicated and difficult to implement, with many opportunities for service providers to offer implementation services, customer education, standard setting, and other value-added facilitations. Some of the many types of business models that have developed with enterprise software and cloud computing might be applicable, too, for the Bitcoin economy—

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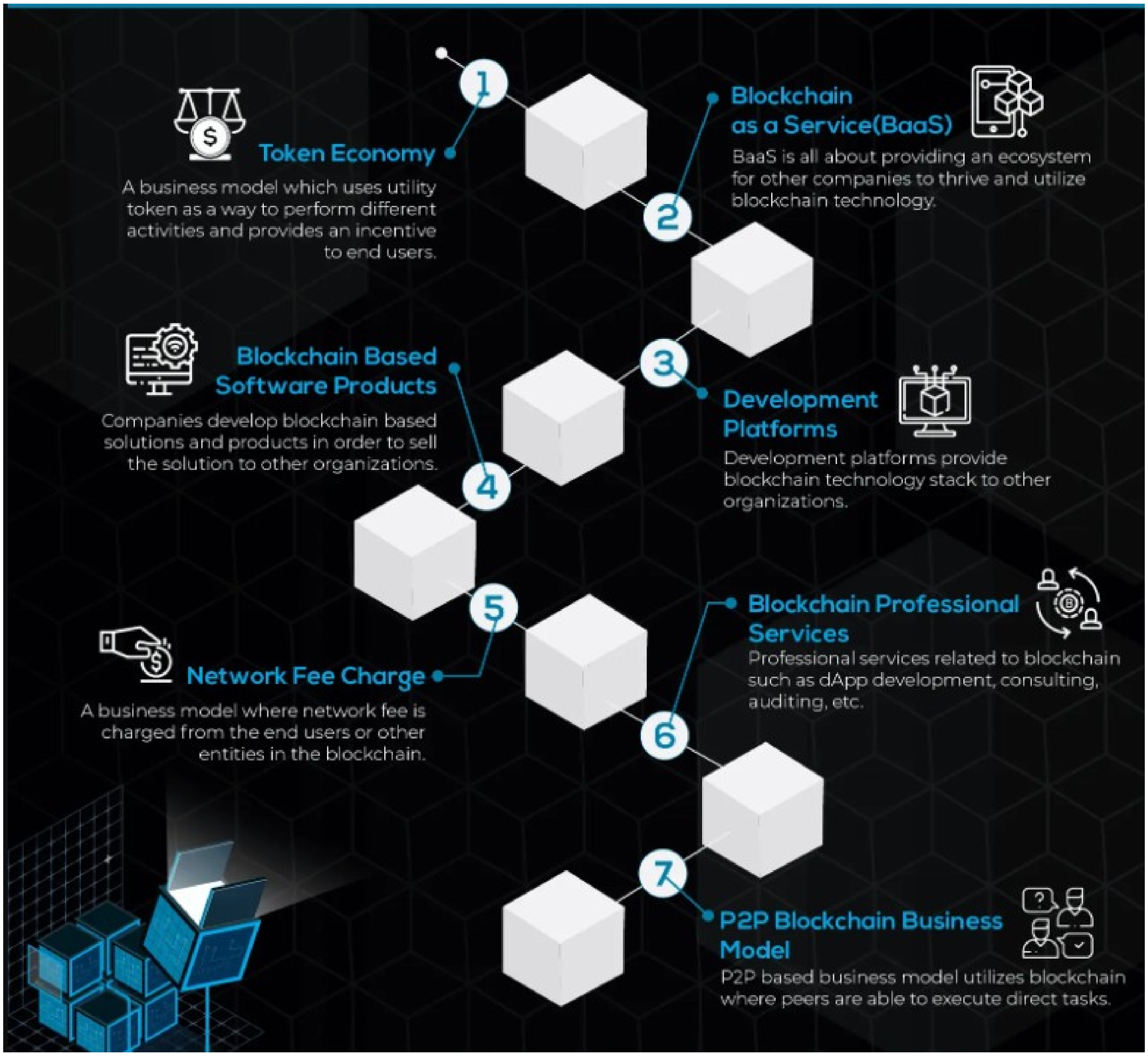
for example, the Red Hat model (feebased services to implement open source software), and SaaS, providing Software as a Service, including with customization. One possible job of the future could be smart contract auditor, to confirm that AI smart contracts running on the blockchain are indeed doing as instructed, and determining and measuring how the smart contracts have self-rewritten to maximize the issuing agent's utility.



Scandals and Public Perception: One of the biggest barriers to further Bitcoin adoption is its public perception as a venue for (and possible abettor of) the dark net's money-laundering, drugrelated, and other illicit activity—for example, illegal goods online marketplaces such as Silk Road. Bitcoin and the blockchain are themselves neutral, as any technology, and are "dual use"; that is, they can be used for good or evil. Although there are possibilities for malicious use of the blockchain, the potential benefits greatly outweigh the potential downsides. Over time, public perception can change as more individuals themselves have ewallets and begin to

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use Bitcoin. Still, it must be acknowledged that Bitcoin as a pseudonymous enabler can be used to facilitate illegal and malicious activities, and this invites in-kind "Red Queen" responses (context-specific evolutionary arms races) appropriate to the blockchain. Computer virus detection software arose in response to computer viruses; and so far some features of the same constitutive technologies of Bitcoin (like Tor, a free and open software network) have been deployed back into detecting malicious players. Another significant barrier to Bitcoin adoption is the ongoing theft, scandals, and scams (like so-called new altcoin "pump and dump" scams that try to bid up new altcoins to quickly profit) in the industry. The collapse of the largest Bitcoin exchange at the time, Tokyo-based MtGox, in March 2014 came to wide public attention. An explanation is still needed for the confusing irony that somehow in the blockchain, the world's most public transparent ledger, coins can disappear and still remain lost months later. The company said it had been hacked, and that the fraud was a result of a problem known as a "transaction malleability bug." The bug allowed malicious users to double-spend, transferring Bitcoins into their accounts while making MtGox think the transfer had failed and thus repeat the transactions, in effect transferring the value twice.

Blockchain industry models need to solidify and mature such that there are better safeguards in place to stabilize the industry and allow both insiders and outsiders to distinguish between good and bad players. Oversight need not come from outside; congruently decentralized vetting, confirmation, and monitoring systems within the ecosystem could be established. An analogy from citizen science is realizing that oversight functions are still important, and reinforce the system by providing checks and balances.

Government Regulation: How government regulation unfolds could be one of the most significant factors and risks in whether the blockchain industry will flourish into a mature financial services industry. In the United States, there could be federal-and state-level legislation; deliberations continue into a second comment period regarding a much-discussed New York Bitlicense.

The New York Bitlicense could set the tone for worldwide regulation. On one hand, the Bitcoin industry is concerned about the extremely broad, wide-reaching, and extraterritorial language of the license as currently proposed. The license would encompass anyone doing anything with anyone else's Bitcoins, including basic wallet software (like the QT wallet).

However, on the other hand, regulated consumer protections for Bitcoin industry participants, like KYC (know your customer) requirements for money service businesses (MSBs), could hasten the mainstream development of the industry and eradicate consumer worry of the hacking raids that seem to plague the industry.

There could be an overhaul of the taxation system to a consumption-based tax on large-ticket visible items such as hard assets (cars, houses). Chokehold points would need to be easily visible for taxation, a "tax on sight" concept. A potential shift from an income tax–based

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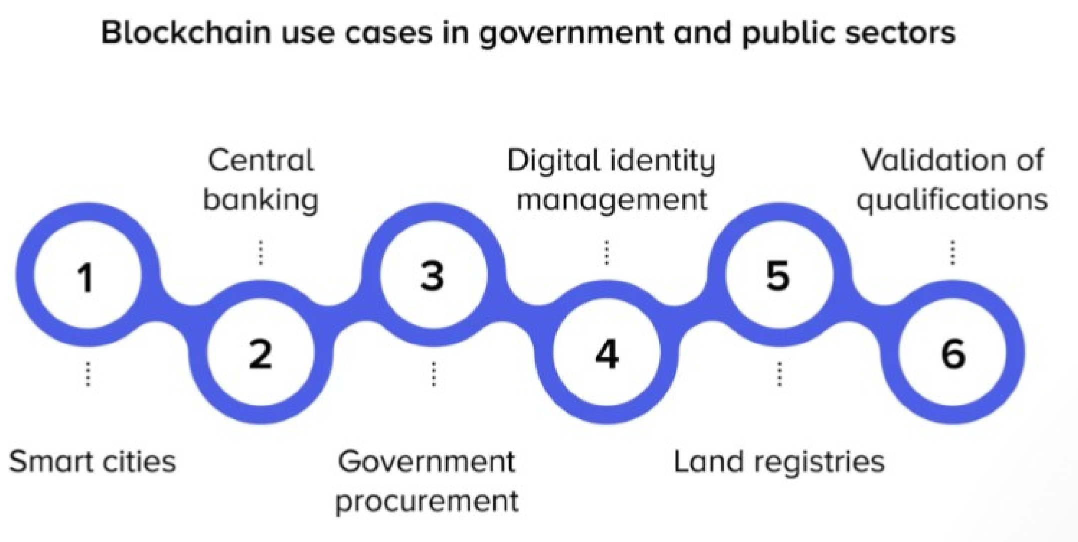
system to a consumption tax–based system could be a significant change for societies. A second issue that blockchain technology raises with regard to government regulation is the value proposition offered by governments and their business model. Some argue that in the modern era of big data, governments are increasingly unable to keep up with their record-keeping duties of recording and archiving information and making data easily accessible. On this view, governments could become obsolete because they cannot fund themselves the traditional way—by raising taxes. Blockchain technology could potentially help solve both of these challenges, and could at minimum supplement and help governments do their own jobs better, eventually making classes of governmentprovided services redundant. Recording all of a society's records on the blockchain could obviate the need for entire classes of public service. This view starkly paints governments as becoming redundant with the democratization of government features of the blockchain.



Uses of Block chain in E-Governance:

Block chain is a decentralized, distributed digital ledger that issued to record transactions across a network of computers. It uses cryptography to ensure the security and immutability of the recorded data.

In a block chain, transactions are grouped into blocks and each block is linked to the previous block, forming a chain of blocks. Once a block is added to the chain, it is very difficult to alter the information contained within it. This makes block chain technology particularly useful for storing and managing data that needs to be reliable and secure, such as financial transactions or legal documents.

Block chain technology has the potential to disrupt a wide range of industries, including finance, supply chain management, and government. It is also being explored for use in a variety of other applications, such as voting systems and identity verification.

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One of the main benefits of block chain technology is that it allows for the secure and transparent exchange of information and value without the need for a central authority or intermediary. This can help to reduce costs, improve efficiency, and increase trust among participants.

Block chain technology has the potential to revolutionize the way that governments operate and interact with citizens. Some potential applications of block chain in e-government include:

Secure record-keeping: Block chain can be used to securely store and manage government records, such as birth certificates, land titles, and voting records. This can help to reduce fraud and improve the efficiency of government operations.

Digital identity: Governments can use block chain to create secure, verifiable digital identities for citizens. This can facilitate a range of e-government services, including online voting, tax filing, and access to social services.

Contract management: Governments can use smart contracts to automate the execution of contracts, reducing the need for manual processes and improving efficiency.

Public service delivery: Block chain can be used to deliver public services more efficiently and transparently, such as through the use of decentralized applications(dApps).

Land registry: Many countries have complex and outdated systems for recording and transferring land titles, which can be prone to fraud and corruption. Blockchain can be used to create a secure and transparent land registry, ensuring that property rights are accurately recorded and protected.

Supply chain management: Governments often rely on complex supply chains to deliver goods and services to citizens. Block chain can be used to track and verify the movement of goods through these supply chains, improving transparency and reducing the risk of fraud and corruption. Overall, the use of block chain in e-government can help to improve the transparency, security, and efficiency of government operations, and can provide citizens with greater access to government services.

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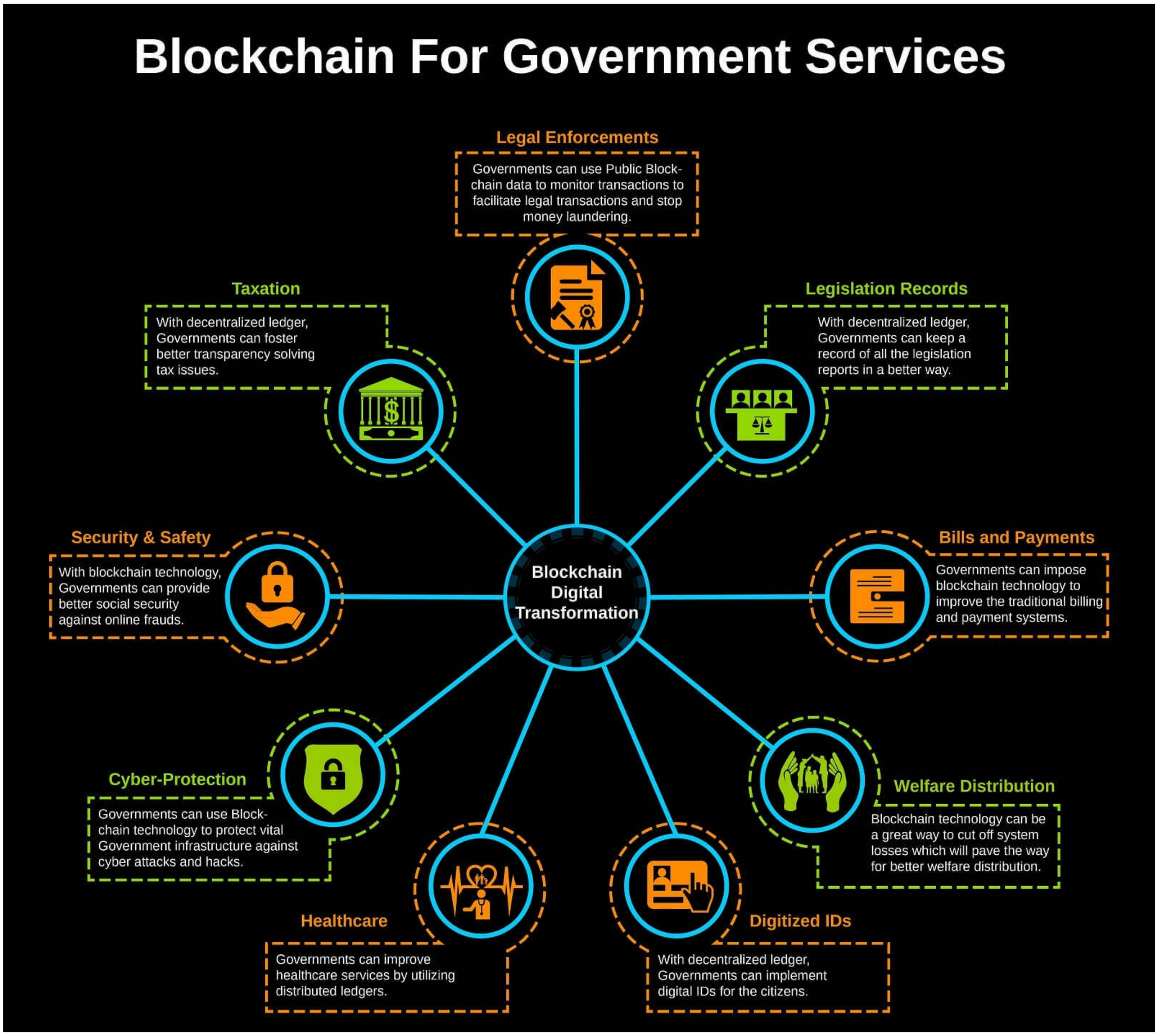

Uses of Block chain in Land Registration:

Stakeholders involved in the Blockchain Land Registry Platform:

1. **Buyer:** A person who buys the land and uses the platform to search the property, request access and interact with the seller and get the land title ownership.
2. **Seller:** A person who sells the land and uses the platform to manage properties and transfer land title to buyers.

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3. **Land inspector:** A person who uses the platform to manage property requests, view reports, confirm and initiate the transfer.

Step 1: Users register to the platform

Users who either want to sell or buy properties register to the blockchain land registry platform.

They can create the profile on the platform with details like name, government-issued ID proofs and designation. A hash for the identity information submitted by the users gets stored on the blockchain.

Step 2: Sellers upload the property specifications on the platform

Sellers can upload properties' images and documents on the platform and pin the land's location on the map. The transaction corresponding to the seller's action of listing the property details is recorded on the blockchain.

Once the property's details are uploaded to the platform, it is made available to all users who have signed up as a buyer.

Step 3: Buyers request access to the listed property

A buyer interested in any specific property can send a request to access its specification to the seller.

Sellers receive notification for property access requests. They can either deny or accept it by looking at the buyer's profile.

Buyers can view the previous ownership records of the property and send a request to purchase it and initiate the transfer.

Transactions corresponding to the requests made by both sellers and buyers are recorded on the blockchain to ensure authenticity and traceability.

Step 4: Sellers approve the transfer request and land inspector gets the notification

If the seller approves the land ownership transfer request, the land inspector gets the notification to initiate the transfer of property. Smart contracts trigger to provide land documents' access to the land inspector.

After the land inspector verifies the documents, they schedule the meeting for ownership transfer with buyer and seller.

The meeting record is also added to the blockchain to solve property related disputes if occur in the future.

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Step 5: Land Inspector verifies the transaction and initiates the transfer

Land inspector verifies the documents submitted by buyers and sellers and adds the authenticated records to the **blockchain land registry** platform.

Sellers and buyers sign the property ownership transfer document in front of the land inspector on the land registry platform.

The signed document gets saved in the database and transaction corresponding to it is recorded on the blockchain.

The transfer is initiated and smart contracts trigger to send funds to the seller and title's ownership to a new buyer.

Step 6: Land Registry Document Validation and Authenticity

In case of any disputes, any authorized party can upload the signed land registry document on the platform to check its authenticity and validate it.

If hash generated after uploading the document is the same as that of the hash created at the time of signing the document, then the document is authenticated and no modifications have been made to the document.

Challenges in the existing land registry process

- **The Involvement of middlemen and brokers**
  Middlemen and brokers are an integral part of every big business as they know more about market offerings. Buyers and Sellers usually prefer to call them to build a full support team.As a result, buyers acquire a deeper understanding of the market and identify lower/higher prices for the transaction. Middlemen gather required information from traders, identify errors, interpret and facilitate the implementation of real estate transactions.Since real estate is big business, it involves a huge number of players, including brokers, lenders, intermediaries and local governments. It leads to additional costs, making the entire ecosystem expensive.

- **The increasing number of fraud cases**
  There has been several cases of imposters posing as the seller of a property. If an imposter successfully pretends as a property owner, they may receive the full amount of after completion and escape with the funds.In many of the cases, both sellers and buyers were unaware of the fraud until discovered by the land registry as part of a spot check exercise.

- **Time Delays**
  Land Registry takes a considerably long time to complete title registrations. There could be a gap of several months between completion and registration.Many legal

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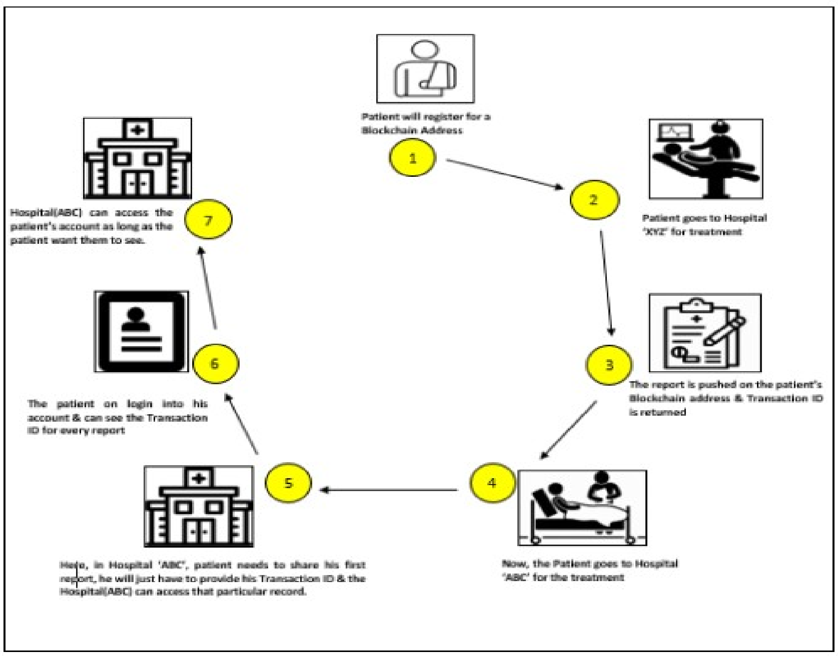
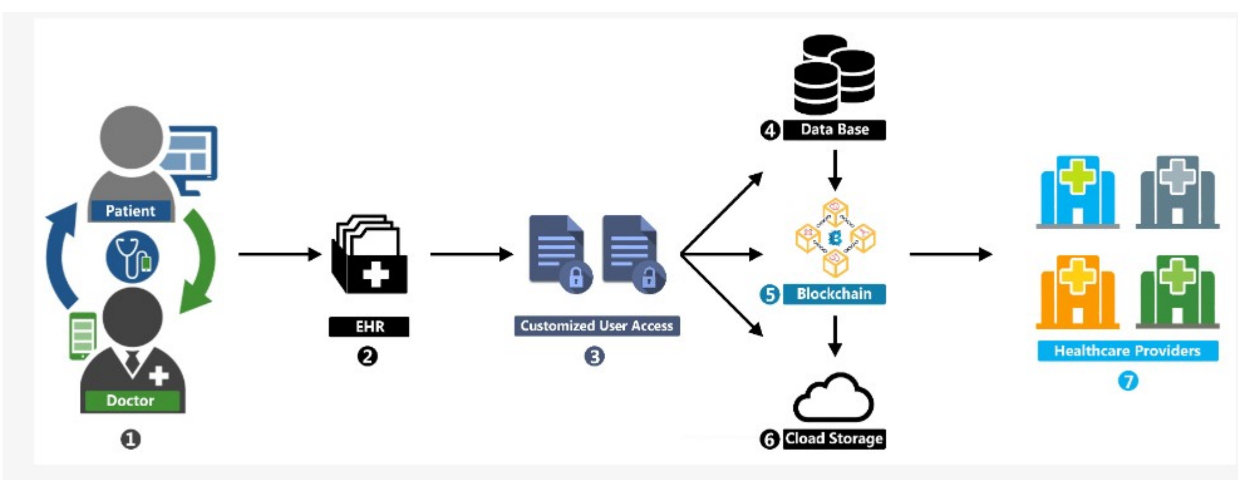
problems can also arise during this long gap. For example, what if you have to serve the landlord's notice to break a lease where property has been sold. Such issues can make the entire process delayed and buyers have to wait for a long time.

- **Human error/intervention**
  Currently, updates to the land registry records are made manually and the accuracy of those changes depends on a particular individual. It means that the land registry is more vulnerable to human errors.Human intervention can increase the chances of errors in the land registry system.

Uses of Block chain in Medical Information Systems:





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A blockchain is a chain of blocks linked together using hashing technique. Each block consists of some timestamped records of information such as financial, healthcare, confidential data, etc. The Blockchain network is managed by a group of users on a decentralized network. All the information is available on a distributed ledger to all the users for transparency. The important and confidential data is encrypted using cryptographic algorithms such as RSA-256. Blockchain is the most hyped technology of Internet 4.0 as it has enhanced security over the internet drastically. Using blockchain on your application or services, security breaching can be reduced by 99% because nothing is 100% secure. The world of computers is working on binary digits and that can be hacked after some decoding but blockchain encodes the data in a way that is not practical to decode easily. In this article, we will discuss the benefits of blockchain in the healthcare sector.

Why Blockchain for Healthcare Sector?

Blockchain is a distributed ledger system that records transactions in a database made up of blocks where blocks are chained together with the hash value to form a chain-like structure. Blockchain has many features that make them a good choice for the healthcare systems:

- **Secure:** Blockchain is a secure network of blocks. Once a block is added to the blockchain, it will require the consensus of the majority of other nodes in the network to alter the data in the block.
- **Decentralized:** Blockchain is a decentralized network of nodes where no one has to trust or know anyone else in the network. Each member in the network has exact same copy of data as other nodes in the network. If a member's ledger is altered, then it will be rejected by the majority of the member nodes in the network.
- **Self-correcting:** When the data in one of the blocks in the network is changed then other nodes in the network check their blocks, and cross-reference each other to root out the error.
- **Distributed:** Blockchain is a distributed ledger where independent computers record, share, and synchronize the transactions in their respective electronic ledgers.

Benefits of Blockchain in the Healthcare Sector

1. Securing Patient Data: In the healthcare sector, securing patient data is the most critical part. Tampering the patient records can create problems to identify the disease or problem in the patient by the doctors and hospitals. Between the years 2009 and 2017, around 176+ million patient records were breached. The hackers stole the credit card and bank information and used them in unethical ways. Blockchain is a tamper-proof, distributed, and incorruptible helps in accessing patient data in a much easier and more secure way.

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2. Medical Drugs Supply Chain Management: Medicines or drugs are not made in hospitals. They are made in labs and pharmaceutical companies around the world. These drugs are further supplied across the countries according to their needs. What happens if someone tampers the drugs while transporting them across the country? So, the medical supply chain has to be tamper-proof and transparent for the importers and exporters. Blockchain fades away this problem as blockchain has features such as transparency, decentralization, and tamper-proof. Once a distributed ledger is created for the drug, each transporting point will be entered into the blockchain from point of origin to destination making it transparent across the transportation.

3. Single Longitudinal Patient Records: As blockchain is a chain of blocks, every type of patient record will be entered into the blockchain ledger. Records such as disease history, lab test results, treatment fees, etc. can help in forecasting the disease based on his next visit using Machine Learning and Artificial Intelligence. Using these precompiled records helps hospitals in offering discounts to their customers by analyzing these records. Also, it will help in mastering patient indices making records streamline carefully, and preventing costly mistakes.

4. Supply chain optimization: Primary challenge in the healthcare sector is to ensure the authenticity of the origin of medical goods to ensure the authenticity of medicines. With the help of blockchain technology, the foods can be traced from manufacturing to every stage of the supply chain. This enables the complete visibility, and transparency of the goods to buy. This can help companies to apply AI and predict demand better and optimize the supply accordingly and at the same time, it can increase the confidence of the customers.

5. Drug Traceability: Blockchain is the most reliable, dependable, secure solution to track every drug to its roots. Every block of data containing drug information will have a hash value linked to another block of data. This hash value ensures that the data has not been tampered with. The transactions in the blockchain are visible to all authorized parties. Medicine buyers will be able to ensure the authenticity of the purchased products by scanning the QR code and looking up all the necessary information like manufacturer's detail, etc.

6. Cryptocurrency Payments: Blockchain makes it possible to receive medical assistance and pay for those services in cryptocurrency. For example, Aveon Health and Micropayment are the technology that is based on Blockchain and uses the advantages of Blockchain technology. Aveon Health uses the advantages of Bitcoin cryptocurrency. The micropayment model will record every detail of the patient's activities related to the treatment reexamination.

7. Decentralized storage of medical records: The interplanetary file system (IPFS) enables storing versioned file data in decentralized storage environments. The filesystem creates a single hash called base configured IED description which ensures that all the versions of the files are identical. The hash returned by the IPFS allows users to access data on websites, but because blockchain files are immutable and websites changes regularly and DNS would need

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to be changed after each update. Thus, IPFS solves this problem by mapping site addresses to DNS.

8. Update medical supply chain management: Blockchain’s security, reliability, and decentralized storage make it well-suited for the managing, and monitoring of the movement of drug supplies. It helps to create a trusted network of vendors to enhance patient safety. Blockchain integrates all the processes like production, packaging, distribution, transportation, and warehouse information in a single immutable record that is stored securely.

9. Improves electronic health record systems: Electronic health record systems are the health data in digital format that is created and stored by multiple healthcare facilities. Blockchain addresses shortcomings like accessibility, interoperability, and authentication by linking electronic health records and sharing ownership of the records among all stakeholders.

10. Improved recruitment for clinical trials: Researchers have created an Ethereum-based blockchain that simulates the recruitment process. The resulting blockchain-based system allows all researchers to view trial information while protecting the privacy of the trial participants.

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